MICROSOFT FABRIC PROJECT

Supply Chain Optimization & Shipment Tracking Warehouse

Project mission

Build an end-to-end logistics analytics solution using Microsoft Fabric while preserving the supplied datasets, business scenario, SCD Type-2 requirements, SLA logic, and expected outcomes.

Domain	Logistics and Supply Chain
Primary platform	Microsoft Fabric
Team mode	Group implementation and demonstration

1. Project Overview

1.1 Problem Statement

A global logistics provider wants to centralize data from shipments, suppliers, and distribution centers to improve operational visibility and delivery performance. The solution must support historical analysis of supplier and distribution-center changes and provide near real-time logistics KPIs for decision-making.

1.2 Business Objectives

- Optimize delivery performance across routes.
- Track historical changes in supplier information and shipment status.
- Monitor delivery SLA compliance and delay trends.
- Analyze supplier reliability and distribution-center capacity over time.
- Provide a governed, reusable data foundation for reporting and self-service analytics.

1.3 Scope and Success Criteria

Area	Success criterion
Ingestion	All three source datasets are loaded into OneLake through Fabric Data Factory pipelines.
Quality	Nulls, data types, junk values, format consistency, and status codes are validated.
History	Supplier and distribution-center attributes are versioned using SCD Type-2.
Model	A star schema exposes fact_shipments and the required dimensions.
Analytics	A Power BI report displays SLA, delay, reliability, and capacity KPIs.
Operations	Pipelines are monitored, rerunnable, parameterized, and auditable through control metadata.

Important constraint

Use the dataset names and source columns exactly as specified in this document. Additional technical columns may be added only for data quality, lineage, surrogate keys, SCD Type-2, and audit control.

2. Source Datasets

Dataset	Columns
shipments.csv	shipment_id, origin, destination, dispatch_date, delivery_date, carrier_id, status
suppliers.csv	supplier_id, supplier_name, contact_info, region, rating, updated_date
distribution_centers.csv	dc_id, location, capacity, region, updated_date

2.1 Data Rules to Validate

- Primary business keys must not be null: shipment_id, supplier_id, and dc_id.
- Dates must be valid and cast to the expected date or timestamp type.
- delivery_date cannot precede dispatch_date when both values are present.
- rating and capacity must be numeric and within an agreed business range.
- status values must be standardized to an approved list.
- Contact and location fields must be normalized without changing their business meaning.

- Duplicate records must be quarantined or resolved using documented rules.

3. Target Solution in Microsoft Fabric

3.1 Platform Components

Capability	Microsoft Fabric implementation
Unified storage	OneLake with a Lakehouse for Files and Delta tables
Data ingestion	Data Factory pipelines and Copy activity
Transformation	Fabric notebooks using Spark and Delta Lake operations
Data warehouse	Fabric Warehouse for the dimensional serving layer
Control metadata	Fabric Warehouse control and audit tables
Reporting	Power BI semantic model and interactive report
Orchestration	Data Factory pipeline activities, parameters, dependencies, and schedules
Security	Workspace roles, item permissions, OneLake security, and approved connections
Monitoring	Fabric Monitoring hub, pipeline run history, and audit tables

3.2 Logical Architecture

SOURCE	INGEST	LAND	CURATE	SERVE	ANALYZE
CSV files / SQL Server	Fabric Data Factory	OneLake Lakehouse Files	Notebooks + Delta tables	Fabric Warehouse	Power BI

Data flow: source data is ingested to the Lakehouse, validated and versioned in Delta tables, loaded into the Warehouse, and consumed through Power BI.

3.3 Lakehouse Layering

Layer	Purpose	Example objects
Landing	Immutable source-aligned files with ingestion metadata.	Files/landing/shipments, suppliers, distribution_centers
Validated	Cleaned, typed, standardized Delta tables.	validated_shipments, validated_suppliers, validated_distribution_centers
Curated	Business-ready dimensions and facts with history.	dim_suppliers, dim_distribution_centers, fact_shipments
Serving	Warehouse tables/views optimized for BI.	dbo.dim_suppliers, dbo.dim_distribution_centers, dbo.fact_shipments

Design principle

OneLake remains the shared data foundation. The Lakehouse supports engineering and Delta processing, while the Warehouse provides a governed SQL serving layer for reporting.

4. Dimensional Model and Business Logic

4.1 Target Star Schema

Table	Key fields	Purpose
dim_suppliers	supplier_key, supplier_id	Stores supplier attributes and historical versions.
dim_distribution_centers	dc_key, dc_id	Stores location, capacity, region, and historical versions.
fact_shipments	shipment_id, supplier_key, center_key	Stores shipment activity and derived SLA measures.
scd_control	table_name, hashes, status, timestamps	Tracks SCD processing and operational metadata.

4.2 SCD Type-2 Technical Columns

- Surrogate key: supplier_key or dc_key
- Business key: supplier_id or dc_id
- effective_start_date
- effective_end_date
- is_current
- record_hash
- created_timestamp and updated_timestamp
- pipeline_run_id or batch_id

4.3 SCD Type-2 Change Logic

1. Build a deterministic hash from tracked attributes.
2. Match the incoming record to the current target row using the business key.
3. If no current row exists, insert a new active version.
4. If the hash changes, expire the current version and insert a new active version.
5. If the hash is unchanged, do not create another version.
6. Ensure only one row per business key has is_current = 1.

4.4 Tracked Attributes

Dimension	Tracked attributes
dim_suppliers	contact_info, rating, region
dim_distribution_centers	capacity, region

4.5 Shipment SLA Metrics

Core derivations

delivery_delay_days = DATEDIFF(day, dispatch_date, delivery_date)

sla_met = 1 when delivery_delay_days <= 2; otherwise 0. Define treatment for in-transit or missing delivery dates in the solution documentation.

4.6 Recommended Additional Metrics

- On-time delivery percentage

- Average and median delivery delay
- Delayed shipment count and trend
- Average delay by origin, destination, route, and region
- Supplier reliability over time
- Distribution-center capacity trend
- Shipment status distribution
- Data freshness and failed-record count

5. Control Framework

5.1 Control Table

Create the following logical control structure in the Fabric Warehouse. The team may add run-level fields while retaining the named fields below.

Column	Suggested type	Description
table_name	VARCHAR(128)	Target dimension being processed
source_hash	VARCHAR(128)	Hash or aggregate signature from the source batch
target_hash	VARCHAR(128)	Hash or aggregate signature from the current target state
is_active	BIT	Indicates an active control record
start_date	DATETIME2	Processing validity start
end_date	DATETIME2	Processing validity end
last_updated	DATETIME2	Most recent update timestamp

5.2 Recommended Run Audit Table

Field	Purpose
pipeline_run_id	Uniquely identifies a pipeline execution.
source_name	Identifies the input dataset.
load_start_time / load_end_time	Measures runtime and freshness.
rows_read / rows_written	Supports reconciliation.
rows_rejected	Measures data quality failures.
run_status	Succeeded, failed, or partially completed.
error_message	Stores controlled diagnostic details.
watermark_value	Supports incremental processing.

5.3 Data Quality Handling

- Write invalid records to a quarantine Delta table with the rule name and rejection reason.
- Do not silently delete failed records.
- Stop the pipeline for critical key or schema failures; continue with warning thresholds only when documented.
- Reconcile source, accepted, rejected, and target row counts for every run.
- Design reruns to avoid duplicate facts or duplicate current dimension rows.

6. Five-Day Implementation Plan

Day 1: Workspace, Lakehouse, and Ingestion

Objective

Set up the Fabric workspace and build repeatable ingestion pipelines.

- Create a Fabric workspace and assign the required team roles.
- Create a Lakehouse and landing folder structure in OneLake.
- Configure approved connections to SQL Server or file sources.
- Build parameterized Data Factory pipelines for shipments.csv, suppliers.csv, and distribution_centers.csv.
- Store source-aligned data in the landing area and capture ingestion metadata.
- Execute the pipelines and validate file counts and row counts.

Deliverables: functional ingestion pipeline, source data available in OneLake, and initial run evidence.

Day 2: Data Quality and Delta Structuring

Objective

Clean, validate, standardize, and persist the supplied data as Delta tables.

- Use Fabric notebooks for null checks, type casting, masking where required, and junk-value handling.
- Normalize supplier contact values and distribution-center location formats.
- Standardize shipment status values.
- Implement duplicate detection and quarantine handling.
- Write validated datasets as managed Delta tables in the Lakehouse.
- Document quality rules, rejected rows, and reconciliation results.

Deliverables: validated Delta tables, quarantine outputs, and a data-quality summary.

Day 3: SCD Type-2 and Control Metadata

Objective

Enable reliable historical tracking for the two dimensions.

- Create dim_suppliers and dim_distribution_centers with surrogate keys and SCD technical columns.
- Create scd_control and the recommended run audit table in the Fabric Warehouse.
- Implement hash-based change detection in a notebook or supported SQL pattern.
- Expire changed current versions and insert new active versions.
- Load unchanged rows without creating duplicates.
- Test new, changed, unchanged, late-arriving, and rerun scenarios.

Deliverables: SCD Type-2 dimensions, populated control metadata, and test evidence.

6. Five-Day Implementation Plan (continued)

Day 4: Fact Table, SLA Metrics, and Warehouse

	Objective Build the analytical model and serve it through the Fabric Warehouse.

- Create fact_shipments using the supplied shipment columns and required dimension references.
- Derive delivery_delay_days and sla_met using the stated rules.
- Resolve dimension keys using the applicable historical version for the shipment date.
- Load dimensions and facts into the Warehouse.
- Create reporting views and validate referential integrity.
- Review query performance, table distribution, and workload behavior using available Fabric capabilities.

Deliverables: final warehouse schema, populated fact and dimension tables, validated SLA metrics.

Day 5: Power BI, Security, Monitoring, and Handover

	Objective Deliver business insights and complete operational readiness.

- Create a semantic model using the Fabric Warehouse or Direct Lake where appropriate.
- Build dashboards for on-time delivery, average delay by region, supplier reliability, and capacity trends.
- Configure workspace roles, item permissions, and row or column restrictions where required.
- Review pipeline runs in the Monitoring hub and capture operational screenshots.
- Finalize architecture, lineage, data dictionary, SCD logic, test results, and known limitations.
- Conduct a team demonstration and explain design choices.

Deliverables: interactive Power BI report, secured solution, monitoring evidence, and project documentation.

7. Required Power BI Pages

Page	Minimum visuals
Executive Overview	On-time delivery %, total shipments, average delay, status distribution, trend by month
Route Performance	Origin-destination matrix, route volume, average delay, delayed shipment trend
Supplier Reliability	Supplier rating, SLA %, historical attribute analysis, ranking
Capacity Analysis	Distribution-center capacity, regional trend, utilization proxy if defensible
Data Operations	Last refresh, rows processed, rejected rows, pipeline status

8. Testing and Acceptance

8.1 Minimum Test Scenarios

Scenario	Expected result
Initial load	One current version is created for every valid supplier and distribution center.
New business key	A new current dimension row is inserted.
Tracked attribute changes	Old row expires and a new current row is inserted.
Unchanged record	No additional dimension version is created.
Duplicate source row	Duplicate is resolved or quarantined according to the rule.
Invalid date or key	Record is quarantined and counted in the audit output.
Pipeline rerun	Target row counts remain correct with no duplicate active versions.
SLA boundary	A two-day delivery has sla_met = 1; a value above two days has sla_met = 0.
Historical lookup	A fact resolves to the dimension version valid on the relevant event date.

9. Team Submission Pack

- Architecture and data-flow diagram
- Fabric item inventory and workspace structure
- Pipeline screenshots and execution evidence
- Notebook or SQL implementation files
- Lakehouse and Warehouse object list
- Data dictionary and transformation rules
- SCD Type-2 design and test evidence
- Power BI report and KPI definitions
- Security and monitoring notes
- Known limitations and improvement backlog

10. Final Outcome

Expected result

A production-oriented supply chain analytics solution in Microsoft Fabric with governed ingestion, validated Delta data, SCD Type-2 history, a dimensional Warehouse, delivery SLA insights, monitoring, and an interactive Power BI experience.